

HATS-2b

R-2237

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATS-2_250114 · created 2026-08-15T01:47:05+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460689.9954 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.047
Transit depth (Rp/Rs)^2	2.66 +/- 1.53 [%]
Orbital Inclination (inc)	88.9 +/- 2.04
Airmass coefficient 1 (a1)	1.014 +/- 0.035
Airmass coefficient 2 (a2)	-0.004 +/- 0.049
Scatter in the residuals of the lightcurve fit is	3.5 %
Best Comparison Star	#2 - [385, 40]
Optimal Method	PSF photometry
Transit Duration (day)	0.0472 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.163 ± 0.047 vs 0.1335 (deviation 0.63σ)
Duration fit vs published	0.0472 d vs 0.0871 d (deviation 8.49σ)
Mid-time O-C	5.2 min
Depth SNR	3.5
Residual scatter	3.5 %

Light curve

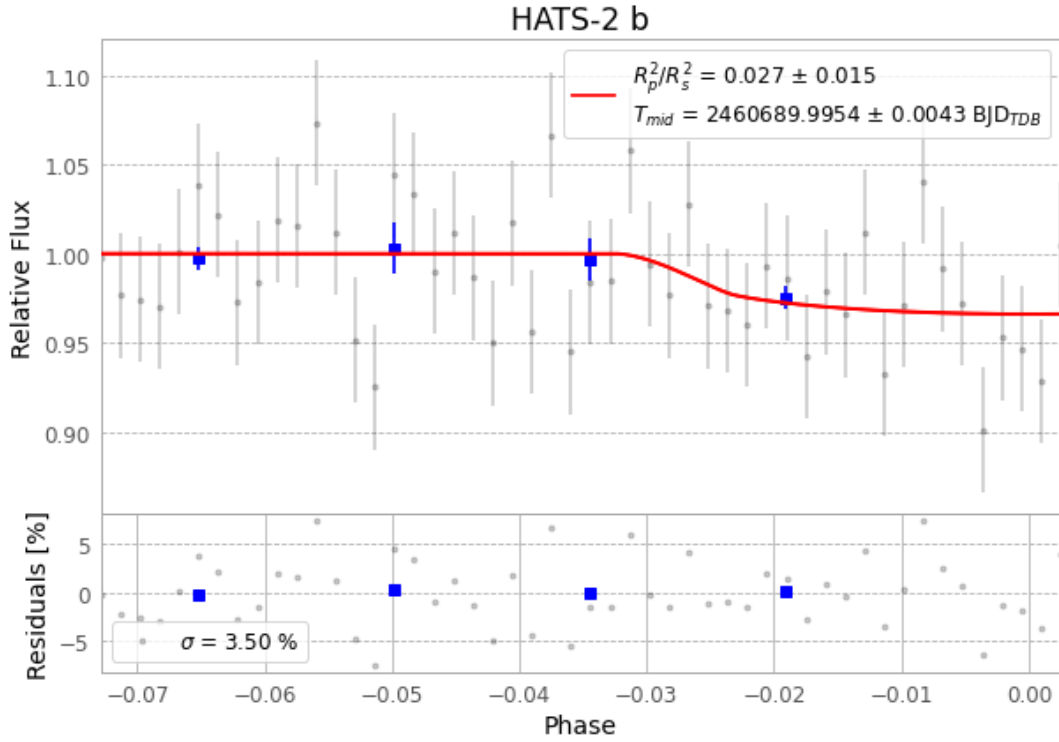


Figure 1 — FinalLightCurve_HATS-2 b_2025-01-14.png (SHA-256 3ab99106fc80c8ad...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [386, 225], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 127035ded50ded99...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

50 FITS frames (33 MB), HATS-2250114092716.FITS → HATS-2250114115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATS-2-250114.log (34f435ffea71...), FinalLightCurve_HATS-2 b_2025-01-14.png (3ab99106fc80...), FinalLightCurve_HATS-2 b_2025-01-14.csv (cff36ebf3e37...)

Compute resources

Wall time 96.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 01:47Z.